HITL Runbook — Edge Detection on the Raspberry Pi 4B

Complete command sequence: host pipeline, Pi edge node, and the ns-3 impaired link. Commands are marked **[HOST]** or **[PI]**.

Architecture

One cable carries two logically separate links, split with 802.1Q VLANs. The camera is treated as the drone's internal camera cable and is never impaired; only detections cross the simulated radio.

```
HOST                                            PI 4B  (UAV2)
Gazebo --- eth-cam  VLAN 10 ==== cable ==== eth0.10 ---> detector
           10.0.0.1                         10.0.0.2     YOLOv8n
                                                            |
gcsns <-- tap-gcs <-- ns-3 <-- tap-uav2 <-- br-uav2 <-------+
10.42.0.10            ^        eth-rf VLAN 42     eth0.42
gcs_receiver     loss, latency, fading            10.42.0.12
```

Stream	Path	Through ns-3?
/uav1/camera/image_raw	host → Pi, VLAN 10	No — plain cable
/uav1/camera/annotated	Pi → host, VLAN 10	No — debug only
/detections/uav1	Pi → gcsns, VLAN 42	YES — measured path

1. Pre-flight [HOST]

```
cd /home/anton/multi_uav_simulation
ip -4 addr show | grep 10.0.0.1        # must return a line
ping -c 2 10.0.0.2                     # Pi must answer
sysctl -n net.core.rmem_max            # must be 536870912
```

If 10.0.0.1 is missing, NetworkManager has taken the adapter:

```
nmcli connection show                  # find the wired profile name
sudo nmcli connection modify "Wired connection 2" \
     ipv4.method manual ipv4.addresses 10.0.0.1/24 \
     ipv4.never-default yes ipv6.method disabled
sudo nmcli connection up "Wired connection 2"
```

! 10.0.0.1 must exist **before** Gazebo starts. Fast DDS 2.6 reads the interface list once at participant creation and has no dynamic detection, so an address added later can never be used — and the failure is silent.

2. Start the host pipeline [HOST]

```
./scripts/netns/run_hitl.sh --no-pi
```

Brings up netns, ns-3, Gazebo, SITL, micro_ros_agent, drone_bridge, the VLANs and br-uav2, then gcs_receiver inside gcsns. Wait for **HITL STACK RUNNING**. Use `--no-pi` when you start the detector by hand.

Option	Effect
--mission	fly uav1_patrol_mission.py automatically
--record	record the camera to a rosbag for offline replay
--debug	detector saves annotated JPEGs (not for timing runs)
--pt	use the PyTorch model instead of NCNN

--conf <v>	detector confidence threshold (default 0.4)

3. Split the Pi's port into two links [PI]

```
ssh anton@10.0.0.2
sudo bash ~/pi_hitl_link.sh
```

Creates eth0.10 (10.0.0.2, camera) and eth0.42 (10.42.0.12, radio).

> **!** Not persistent — plain `ip` commands are lost on reboot, and the Pi falls back to untagged eth0 while the host still expects VLAN 10. Re-run this after every Pi reboot until it is moved into netplan.

4. Verify both links [PI]

```
ping -c 3 10.0.0.1        # camera link   -> expect ~2 ms
ping -c 30 10.42.0.10     # radio link    -> expect ~59 ms, high jitter
```

Measured reference: camera 2.2 ms avg / 0.8 ms jitter; radio min 3.3, avg 59.0, max 158.0 ms, 40.8 ms jitter, 0% loss. Use at least 30 pings — the first packet includes ARP across the simulated channel and skews a short sample badly.

> **!** If the radio link answers as fast as the cable, detections are NOT crossing ns-3 and every latency number will be meaningless.

5. Start the edge detector [PI]

```
source /opt/ros/humble/setup.bash
export FASTRTPS_DEFAULT_PROFILES_FILE=$HOME/uav2_ws/config/fastdds_hitl_eth.xml

~/yolo_env/bin/python ~/yolo_detect_node.py --ros-args \
    -p model_path:=/home/anton/models/yolov8n.pt \
    -p show_window:=False
```

`show_window:=False` is required — the Pi is headless and cv2.imshow has no display. Swap the model path to `yolov8n_ncnn_model` for the NCNN build.

> **!** The Pi's venv is pinned to `numpy==1.26.4` and `opencv-python==4.10.0.84`. Never run `pip install -U` there: NumPy 2.x breaks cv_bridge with *_ARRAY_API not found*, and OpenCV 5 breaks it with *KeyError: 16*.

6. Watch [HOST]

```
export FASTRTPS_DEFAULT_PROFILES_FILE=/home/anton/multi_uav_simulation/config/fastdds_hitl_et
h.xml
source /opt/ros/humble/setup.bash

ros2 topic hz /uav1/camera/image_raw          # must stay steady when the Pi joins
ros2 run rqt_image_view rqt_image_view /uav1/camera/annotated
tail -f /tmp/hitl_gcs.log                     # detections that crossed ns-3
```

Close the annotated view before any timing run — it sends 2.76 MB per frame back over the cable and costs the Pi a redraw per frame.

7. Fly the mission [HOST]

```
sudo ip netns exec gcsns sudo -H -u anton bash -lc '
    source /opt/ros/humble/setup.bash
    source /home/anton/multi_uav_simulation/ros2/install/setup.bash
```

```
    python3 /home/anton/multi_uav_simulation/ros2/uav_controller/uav_controller/uav1_patrol_mission.py
'
```

Must run inside `gcsns`: drone_bridge lives there, and from the root namespace its services are invisible and the script hangs on *Waiting for service /uav1/arm*.

8. Shut down [HOST]

```
# Ctrl+C in the run_hitl.sh terminal tears down everything, or:
bash scripts/netns/kill_all_netns.sh
ssh anton@10.0.0.2 'pkill -f yolo_detect_node'
```

Measured reference values

Metric	Value
Link throughput (iperf3, steady)	~834 Mbps, 0 retransmits
Camera	1280×720 RGB, 2.76 MB/frame
Camera at 20 Hz / 5 Hz	481 Mbps / ~111 Mbps
Inference — PyTorch @ 960×544	2,540 ms (0.39 fps)
Inference — PyTorch @ 640×384	~1,100 ms
Inference — NCNN @ 960×544	~1,000 ms (0.74 fps)
Inference — NCNN @ 640×384	~470 ms (predicted, not measured)
Detection payload	~118 B vs 2.76 MB/frame image
Recall	2–4 of 5 people per frame

Troubleshooting

Symptom	Cause and fix
Camera rate collapses when the Pi joins	A RELIABLE subscriber throttles the publisher. The detector must use BEST_EFFORT / depth 1.
Camera rate exactly 3.144 s apart	A dead RELIABLE subscriber; Gazebo is retransmitting. Restart the pipeline.
/clock not ticking	The sim has stalled. Lockstep means Gazebo only steps when SITL sends servo packets — check SITL, then restart. Always check /clock first: if it is frozen, every downstream rate is meaningless.
Topic listed but no data	Socket buffers. rmem_max must be 536870912 on both machines; the default 208 KB holds ~7% of one 2.76 MB frame.
XMLPARSER Error: realpath failed	FASTRTPS_DEFAULT_PROFILES_FILE points nowhere. Use the absolute path.
Pi unreachable, Destination Host Unreachable	The Pi rebooted and lost its VLANs; it is untagged while the host expects VLAN 10. Re-run pi_hitl_link.sh.
Package 'uav_vision' not found	uav_vision builds into <repo>/install, not <repo>/ros2/install.
Parameter has not been declared	The Pi has stale code. rsync then colcon build --packages-select uav_vision.